Transverse transport and trapping of submerged structures due to water wave refraction

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Waves refract when traveling over water of variable depth, and the theory of radiation stress implies consequent forces imparted to the bottom topography. Such forces may be relevant for deformable boundaries and submerged objects that are free to move, but such wave-structure interactions have yet to be experimentally observed nor utilized in applications. Here we demonstrate such effects at the laboratory scale by introducing experimental techniques for studying the movements of submerged structures due to water waves. Asymmetrically shaped bodies are shown to move perpendicularly to plane traveling waves, and visualizations of the distorted wave fields are consistent with a mechanism based on refraction in the shallow layer over the structure. Further, we show how this effect can be exploited to stably position or "tweeze" objects within focused wave fields. Simplified calculations suggest that refractive redirection of momentum flux can explain these outcomes, and the form of the wave field and geometry of the body can be used to control forces. The systems presented here are hydrodynamic analogs to acoustic and optical transport and trapping, suggesting generalizations that could guide further strategies for wave-based motion control.

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I. INTRODUCTION

The forcing of solid structures and boundaries by water waves is central to many natural phenomena and engineering contexts across widely ranging scales. Studies of such problems have been conventionally motivated by large oceanic structures such as ships, docks, and offshore platforms that are intended to be static or otherwise resist the effects of impacting waves [1–3]. Related natural contexts involve oceanographic topographies and coastlines [4,5]. At smaller scales, mechanical structures may deform or move in response to wave-induced forces, and these motions may also generate waves or otherwise affect the scattered field. Such wave-structure interaction effects have been exploited in energy harvesting devices that involve dynamic components such as point absorbers, attenuators, and terminators [6–8]. At yet smaller scales, there has been growing interest in using water waves for applications such as moving or positioning floating particulates or objects [9–11]. Recent approaches variously rely on the convective forcing induced by wave drift, vortical surface flows, or topological properties of highly structured wave fields. Recent related work has also explored how wave radiation might be exploited for propulsion of small-scale structures [12–15]. Mutual interactions among waves and objects have also been studied down to the millimeter scale in the context of liquid droplets bouncing on a vertically vibrated bath [16–19]. This class of problems has garnered much recent interest for its parallels with quantum mechanical particle-wave systems.

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Such investigations provide useful contexts and motivating problems whose study can improve the understanding of wave-structure interaction mechanisms operating at small scales. While potential flow theory is suitable for pure gravity waves of sufficiently long wavelength [1,20], surface tension effects are increasingly important at the capillary scales of centimeters and smaller [21,22]. Further, capillarity and viscosity have been shown to play important roles in damping high wave numbers and curtailing fluid instabilities ahead of moving bodies [23–25]. Additional complexities arise for dynamic structures that respond on similar timescales as those characterizing the wave motions. Such has been studied for wave damping over boundaries [26,27] and resonances of flexible structures with waves [28]. Experimental studies at laboratory scales provide settings in which such effects can be tuned, controlled, observed, and measured. Recent laboratory-scale work has focused on tailoring the shapes of a static bottom topography in order to achieve wave refraction, guiding, concentration, and cloaking [29–31]. These and related works often exploit parallels with optical, acoustic, and quantum mechanical wave systems [32–34]. While the submerged boundaries used in such studies presumably experience reactive forces associated with their manipulation of the wave field, no previous experiment seems to have sensed or allowed for free motions that could serve as signatures of such forces.

Here we take some first steps by asking if wave-induced forces on submerged objects are detectable at the laboratory scale, what special methods are needed to do so, and what potentially useful outcomes might be achieved. We focus on the challenging problem of harnessing traveling waves to exert transverse or lateral forces that might induce motions of an object perpendicular to the direction of wave propagation. Our scheme aims to pass waves over a suspended body and thereby induce refraction due to the differing depths, and the resulting change in wave momentum flux is associated with a reactive force on the structure [35,36]. To this end, we introduce experimental techniques for suspending objects just below the free surface and isolating free motions transverse to imposed gravocapillary waves. We study two systems that are inspired by analogous strategies for propelling microparticles perpendicularly to a light beam and for stably holding or "tweezing" molecules and particles using so-called optical or acoustic trapping [37–42]. By demonstrating such behavior using laboratory-scale water waves interacting with mobile structures, the findings reported here advance the study of wave-structure interactions and suggest applications for controlling motions using waves.

II. EXPERIMENTAL METHODS

Here, we briefly review the theoretical basis for water wave radiative stresses and estimate an upper bound for the resulting forces achievable at the laboratory scale. We then describe the experimental details that allow for the observation and measurement of the dynamics of submerged bodies resulting from these surface wave refractive forces. Toward this end, we describe the construction of wave tank and imaging systems along with the characterization of the waves used for transport and trapping experiments. Then we describe the specialized surface tension suspension assembly and linear bearing which allow for low-friction translation. Finally, we discuss our techniques for the determination of suspension depth and hydrodynamic drag on the bodies, which are used to calculate estimates of the refractive forces.

A. Radiation stress from water waves at laboratory scales

To guide our experimental design and motivate parameter choices, we first ask what typical scales for wave-induced forces can be expected in the laboratory setting. For this purpose we consider the radiation stress in a wave field as computed in the classical work of Longuet-Higgins and Stewart for the case of plane gravocapillary waves traveling on the surface of a fluid of uniform depth [21]. The derivations therein for the time-averaged and depth-integrated momentum flux in the direction of wave propagation provide expressions in terms of relevant wave and fluid properties. Saving a more complete analysis for a later section, it can be shown that the scale of the corresponding forces

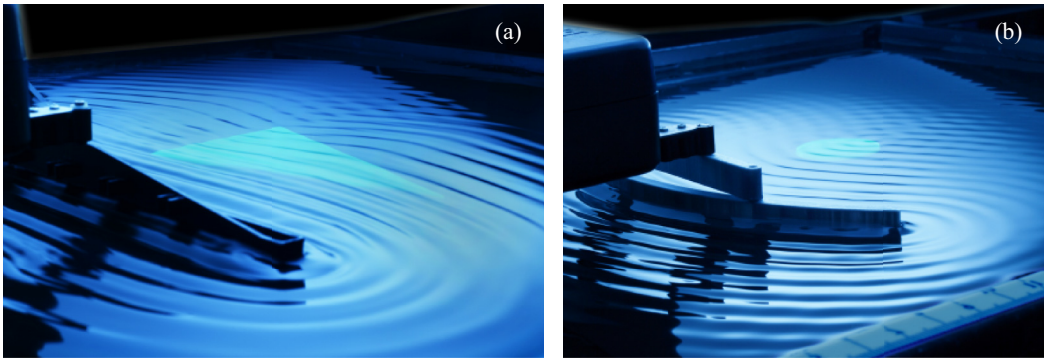


FIG. 1. Experimental systems involving traveling water waves passing over submerged bodies. The photographs are lit with a xenon lamp stroboscope whose flashes are frequency matched to the waves [43]. (a) Triangular body (length 10 cm) immersed under water and interacting with plane traveling waves emitted from a vibrating wavemaker. The waves are refracted as they pass over the body, which acts like a prism. (b) Traveling waves emitted by an arc-shaped wavemaker converge over a disk (diameter 3 cm).

$F \sim \rho g a^2 L$ dominantly depends on the density ρ of the fluid, the gravitational acceleration g, and the amplitude a of waves scattered from an object of typical length L. Considering the known constants $g = 980$ cm/s^2 and $\rho = 1$ g/cm^3 for water, and using the measured values $a \sim 0.01 - 0.1$ mm and $L \sim 10$ cm to be reported in what follows, we arrive at forces of order $F \sim 0.1$ dyn. The corresponding mass equivalent is 0.1 mg, which is about that of a grain of salt. While understood to be rough order-of-magnitude estimates, this prediction usefully shows that the wave-induced forces are expected to be minuscule and hence detecting their effects in the laboratory will be challenging. These considerations motivate the need for special experimental methods with adequate sensitivity.

B. Overview of experimental approach

Our approach exploits the refraction that occurs when water waves travel over spatially changing bottom topographies and thus different depths [44,45]. We propose to induce refraction by passing waves over submerged bodies that are free to move and whose response therefore informs wave-induced forces. Our specific experimental systems, introduced in the photographs of Fig. 1, show traveling waves emanating from vibrating rods and passing over planar objects. The image of (a) captures plane waves over a triangular body and (b) shows circular waves converging over a circular disk. These systems will be shown to demonstrate motions perpendicular to the wave propagation, leading to directed transport in the former and stable trapping in the latter.

Our experiments employ a ripple tank apparatus that allows for the controlled generation of waves and their absorption at the boundaries, as shown schematically in Fig. 2(a) and described in greater detail in Sec. II C. A transect and underwater view of the situation we aim to achieve is shown in Figs. 3(a) and 3(b), respectively: waves emitted by a wavemaker pass over a body, here a fully submerged thin plate of triangular planform, inducing refraction that manifests as a shortening of wavelength over the body. The characterizations of the wave properties are described in Sec. II D.

The key experimental component that enables the results reported here is a means for suspending the body beneath the free surface in a manner that allows nearly frictionless motion lateral to, and under the action of, imposed traveling waves. Isolating this motion along the transverse axis is crucial to resolving the intended effects in an unambiguous way. Our strategy, discussed in depth in Sec. II E, exploits surface tension for both suspension and linear motion via a custom-designed fluidic bearing [46–49]. The resulting depth of the immersed body can be calculated from its geometry, as detailed in Sec. II F. The expression for hydrodynamic drag on the suspension system, given in Sec. II G, provides a mapping between the motion of the body and the propulsive force that must be applied to achieve steady-state motion.

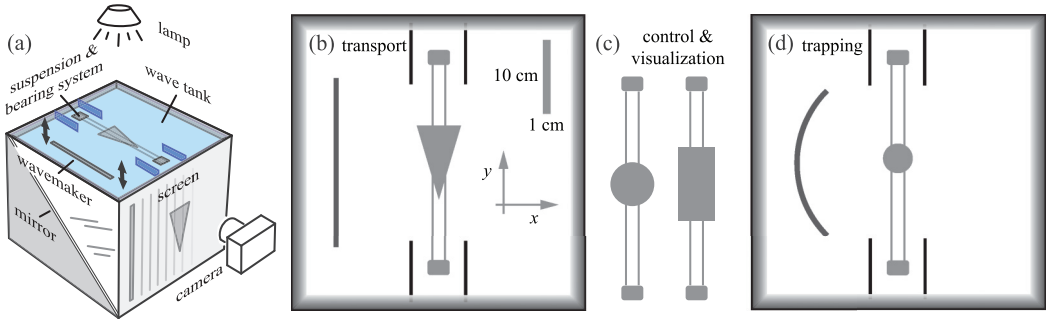


FIG. 2. Experimental apparatus and body assemblies. A ripple tank holds water of depth $H = 0.65$ cm bounded by wave-absorbing "beaches" and a clear glass bottom. Surface waves emitted by a vertically oscillated bar are projected onto a screen by an overhead light. The resulting shadowgraphs may be captured by a camera. A submerged object, here a thin plate of triangular planform, is held via a suspension system that also acts as a linear bearing allowing motion of the object transverse to the direction of propagation of the traveling waves. (b) Components shown at accurate relative scale for the transport experiments. Foam beaches bound the pan, and the rectangular wavemaker is shown to the left. The floater-body assembly sits within four glass panes. A scale bar is shown at the upper right. (c) Disk used as a control in the plane wave experiments and rectangle used to visualize refraction. (d) Circular-arc wavemaker and disk-shaped body used in the trapping experiments.

C. Wave tank apparatus

The wave tank portion of the apparatus is a modification of the ripple tank system offered by the scientific equipment supplier PASCO (product code WA-9896). As shown in Fig. 2(a), the basic components include an elevated pan, a wavemaker consisting of a motor that drives up-and-down

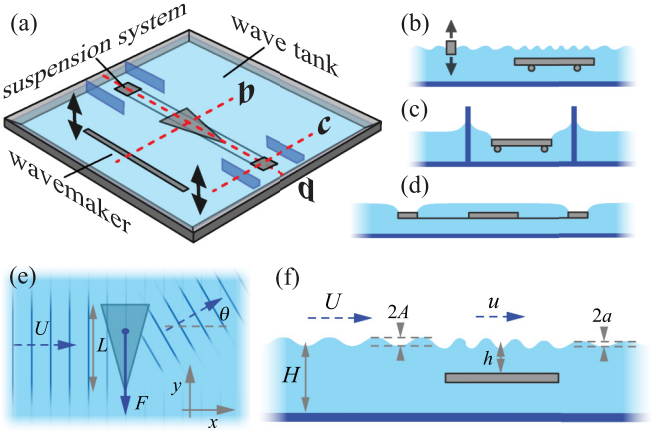


FIG. 3. Transport of asymmetric bodies in plane waves. (a) A ripple tank measuring 40 × 40 cm holds water of depth 0.65 cm bounded by wave-absorbing ”beaches.” A submerged object, here a triangular plate, is held via a suspension system that also acts as a linear bearing allowing motion of the object transverse to the traveling waves. (b) Transect showing the wavemaker and waves that refract in the shallow layer over the object. (c) Floaters connected to the object via thin rods use surface tension to both hold the object just below the free surface and to center the suspension system between vertical glass panes (dark blue). (d) Transect showing the suspension of the central object by two floaters. (e) Overhead view of a triangular plate suspended below a free surface carrying waves along x. The plate refracts waves and is free to move transversely along y. (f) Sideview schematic.

vibrations of a bar, and an overhead lamp that projects shadowgraphs onto a screen via a 45° mirror beneath the glass-bottomed pan. The pan has inner dimensions of 40 × 40 × 1 cm in length, width, and depth. It is bounded on four sides by sloped foam "beaches" that extend 3 cm into the pan and whose purpose is to absorb waves and suppress reflections. For all experiments reported here, we fill the tank to a depth of $H = 0.65$ cm.

A top-view schematic of the transport experiments is given in Fig. 2(b), where the components are shown at accurate relative scale. The wavemaker consists of a motor that drives vertical vibrations of controlled and tunable frequency and amplitude. To generate plane waves, a rectangular-section bar is attached to arms extending from the motor and the bar lowered into the water filling the pan. Vibrations cause the emission of plane traveling waves. The bar used here measures 23 × 0.8 × 0.8 cm in length, width, and depth. The frequency of the motor can be varied via a dial and its value read from a digital display screen. Using a stroboscopic (strobe) light, we validate that the motor frequency faithfully and directly determines the wave frequency f. The amplitude of motor vibration is controlled by another dial whose turn angle is indicated with a scale with values labeled from 0 to 6. We determine how this "wavemaker amplitude" scale relates to the physical wave amplitude A via a procedure described in Sec. II D. For these experiments we consider wave frequency values of order 10 Hz and wave amplitudes of about 0.5 mm.

A top-view schematic of the trapping experiments is given in Fig. 2(d). To generated focused waves, a bar in the shape of a circular arc is 3D printed and attached to the wavemaker arms. The bar measures 21 × 0.4 × 0.8 cm in arc length, width, and depth and its radius of curvature is 23 cm. When vibrated at the surface, waves of circular fronts are emitted and converge to a focal point at the center of the circle associated with the bar.

The overhead lamp may illuminate the ripple tank in continuous or stroboscopic mode, the latter being useful for "freezing" the wave field and capturing photographs. Light transmitted through the ripple tank is projected onto a screen, and the resulting shadowgraph is photographed with a digital camera (Nikon D610). Surface waves induce refractive bending of the light, which has the effect of making wave crests appear bright and wave troughs dark. The magnification of the projected image on the screen can be tuned by changing the height of the lamp.

D. Wave characterizations

A characterization of the wave parameters is necessary for computing the wave-induced forces on the body as well as how they can be expected to change as these parameters are varied. As relevant to the plane wave experiments, we expect low-amplitude traveling waves for which variations in the height of the free surface take the form $z(x,t) = A\sin(\omega t - kx)$, where x is the direction of wave propagation and t is time. Wave parameters include the frequency f and angular frequency $\omega = 2\pi f$, wavelength λ and wave number $k = 2\pi/\lambda$, and wave amplitude A. In the experiments we directly impose frequency f via the wavemaker, and in this work we explore $f = 8 - 12$ Hz. The resulting wavelength λ and wave number k may then be determined from the dispersion relation for intermediate-depth gravocapillary waves,

$$\omega^2 = \left(gk + \gamma k^3/\rho\right)\tanh(kd), \tag{1}$$

where $g = 980$ cm/s^2 is gravitational acceleration, $\rho = 1$ g/cm^3 is the density of water, and $d = H = 0.65$ cm is the relevant depth of the bulk. The surface tension coefficient of the air-water interface has the nominal value $\gamma = 72$ dyn/cm, which, however, is documented to decrease by as much as 50% over tens of minutes because of surface contamination [50]. In Fig. 4(a), we display experimental measurements (colored markers) of the wavelength λ in the bulk as obtained from shadowgraph images, and the corresponding theoretical prediction is shown as the gray band. Measurement errors are comparable to or less than the marker size, and the upper portion of the theory band corresponds to the nominal value of γ and the lower portion reflects a 50% decrease. The comparison shows that the spread in the measurements could be attributable to surface tension

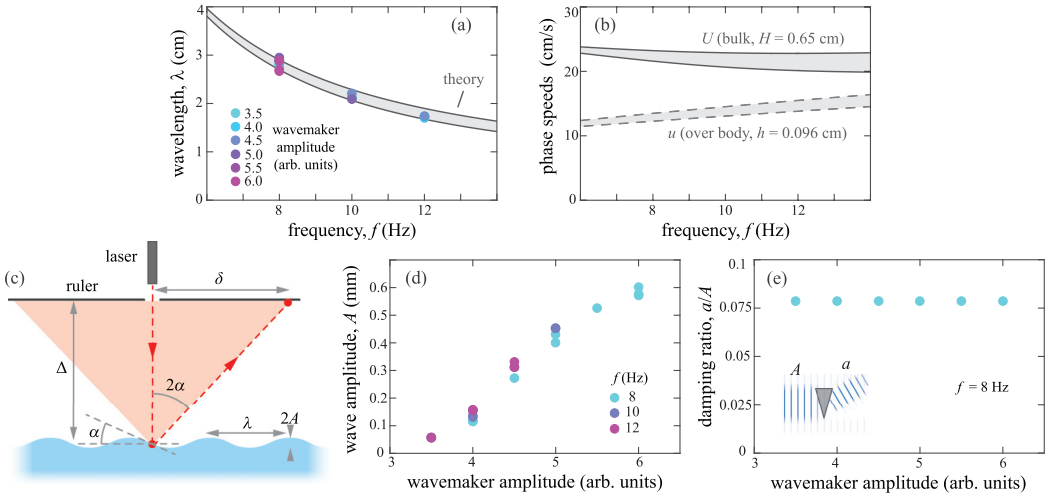


FIG. 4. Wave properties and measurements. (a) Direct measurements (colored markers) of wavelength λ vary with frequency f in a manner consistent with the known dispersion relation (shaded band) for gravocapillary waves of typical surface tension coefficients 72–40 dyn/cm [50]. (b) Wave phase speed as a function of frequency. Curves are shown for depths corresponding to waves in the bulk ($H = 0.65$ cm, solid curves) and in the shallow layer over the triangular body ($h = 0.096$ cm, dashed curves). The shaded bands reflect variations due to surface tension attenuation, as in (a). (c) Determination of wave amplitude, A. Traveling waves cause oscillations of a laser beam that reflects from the surface and is intercepted by an overhead ruler. Measuring the maximal excursion δ informs on the maximal slope of the passing waves, which in turn is related to A and λ. Using a large distance Δ between the ruler and surface ensures that even small A and thus small angle α will yield measurably large excursion δ. (d) Measurements of incoming wave amplitude A show a strong dependence on the wavemaker amplitude and weaker dependence on f. (e) Damping of waves of frequency $f = 8$ Hz passing over a triangular body. The incoming (upstream) amplitude is A and outgoing (downstream) amplitude a, as shown in the inset. The damping ratio takes on values near $a/A = 0.08$ across all tested wavemaker amplitudes.

variations. The typical conditions studied here yield $\lambda = 1 - 3$ cm, and the data confirm that λ depends on f but not the wavemaker amplitude.

The dispersive nature of water waves also implies that the phase speed $c_p = \omega/k$ depends on frequency. Such an effect impacts the refraction that occurs when waves pass over topographies of differing depth. In Fig. 4(b) we display theoretical curves for the faster waves of speed $U = c_p(d = H)$ traveling over the deeper bulk and the slower speed $u = c_p(d = h)$ for the shallow layer over the body, where the reported depth values H and h are relevant to the transport experiments and methods for their determination are to be discussed. The speeds and depths are schematically defined in Figs. 3(e) and 3(f). The bands shown in Fig. 4(b) again reflect surface tension variations. Encouragingly, these predictions indicate that the effective index of refraction $n = U/u$ is significantly large, with typical values in the range of 1 to 2 for the conditions studied here.

The wave amplitudes A are small and necessitate a sensitive measurement technique. We direct laser light downward on the free surface and use its reflection visualized on a ruler above the wave tank to infer amplitude, as shown schematically in Fig. 4(c). As waves travel on the free surface, the location on the ruler where the reflected beam is intercepted is seen to oscillate (rapidly) back and forth. The maximal extent of its excursion informs the steepest slope of the wave, which is related to A. In this way, small amplitudes may be readily determined by simply securing the ruler at a large distance Δ from the free surface, leading to large maximal excursions δ even for small slope angles α. We achieve $\Delta = 2$ m by affixing a ruler to the laboratory ceiling. The maximal slope of the surface is given by $\tan\alpha = 2\pi A/\lambda$ and the maximal excursion by $\delta = \Delta \sin[2\arctan(2\pi A/\lambda)]$.

This leads to a formula for the amplitude A in terms of the measured quantities Δ, δ, and λ: $A = (\lambda/2\pi)\tan[\frac{1}{2}\arcsin(\delta/\Delta)]$. Because the waves may decay in amplitude away from the wavemaker due to radial spreading and viscosity [51], we opt to measure the amplitudes just upstream and downstream of the body at its spanwise midpoint. These locations inform on the incoming waves and those refracted, respectively. In Fig. 4(d) we display the measured values of the amplitude A just upstream of the body and for the entire range of f and wavemaker amplitudes explored in this work. It will prove most convenient to use somewhat different amplitudes for the transport and trapping experiments.

Further, we find that the waves are damped significantly upon passing across the shallow layer over the bodies. This effect is shown schematically in Fig. 3(f). It can be quantified by the ratio a/A, where A is the wave amplitude imposed by the wavemaker just upstream of the body and a is the (smaller) amplitude just downstream of the body. In Fig. 4(e) we show measured values of this damping ratio for conditions relevant to the transport experiments. It can be seen that the triangular body damps the waves such that the outgoing or transmitted amplitude is about 8% of the incoming amplitude. This effect is important in estimating the momentum in the refracted waves and thus the force experienced by the body, as to be shown in the theory of Sec. III B.

For the transport experiments we vary f and fix the wavemaker amplitude at 6, yielding incoming plane waves of amplitude $A = 0.57\,\text{mm}$. For the trapping experiments we set $f = 12\,\text{Hz}$ and a wavemaker amplitude of 4, leading to a wave amplitude of 0.16 mm near the wavemaker. In this latter case, the amplitude increases in the approach to the focal point of the converging waves.

E. Suspension assembly and bearing system

Our experiments exploit surface tension effects to both suspend objects just below the air-water interface and to provide a low-friction linear bearing that permits motion of the objects due their interaction with waves. The objects studied here are made of thin acrylic plastic laser-cut into simple planform shapes such as triangles, disks, and rectangles [Figs. 2(b)–2(d)]. Each body is glued to two thin carbon fiber rods, and two "floaters" are also glued to the ends of the rods. Long rods are used to ensure that the floaters are sufficiently far away from the body so as not to interfere with the local wave field. The floaters are rectangular and cut from acrylic of the same thickness as the body. This ensures that the upper surfaces of the body and floaters are coplanar. The assembly is denser than water and sinks when fully submerged. However, if the assembly is carefully lowered into water with the carbon fiber rods immersed and the upper surface of the body is wetted while the upper surfaces of the floaters are carefully kept dry, the assembly may be made to hang just beneath the air-water interface, as shown in Fig. 3(d). At this point, the body is completely submerged with its upper surface at a (small) depth h below the interface, while the upper surfaces of the floaters remain dry and have downward-bending menisci along their peripheries. The suspension of the assembly results from its total weight being balanced by a combination of buoyancy associated with all its components and the surface tension exerted along the circumferential length of the floaters [52]. In Sec. II F we outline how this force balance allows us to infer the value of h, which is about 1 mm for all assemblies studied here.

The downward menisci surrounding the floaters also serve an important role in the bearing system. On either side of each floater we stand panes of glass that breach the free surface. The glass is freshly cleaned and thus hydrophilic or wetting, yielding upward menisci. The floaters thus repel from the glass due to surface tension interactions—i.e., the "Cheerios effect" [53]—and tend to center themselves between the panes as shown in Fig. 3(c). The system acts as a linear bearing with the panes as stators or guides and the floater-body assembly as the slide that may translate within these guides. Importantly, the repelling surface tension interactions ensure that no part of the sliding assembly ever directly contacts the panes nor any other part of the tank. The assembly freely slides transverse to the direction of wave propagation, only resisted by fluid friction but no solid-solid friction that would require a threshold force for movement. Thus any forcing, however small, gives rise to motion, making the system a highly sensitive force detector.

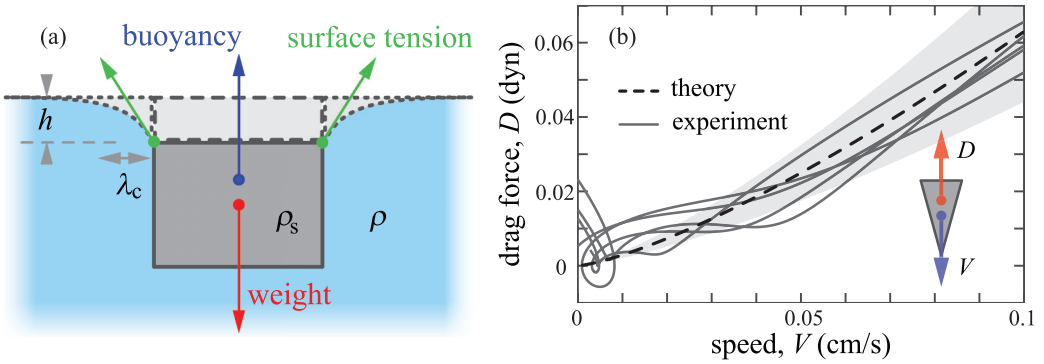


FIG. 5. Inferring body properties. (a) Determination of depth h below the surface. A partially submerged object of density greater than the liquid ($\rho_s > \rho$) sits at a depth such that its weight is balanced by a combination of buoyancy and surface tension. The buoyant force is given by the weight of liquid displaced by the object (dark gray) and the region directly above the object up to the waterline (light gray with dashed outline). The vertical component of the surface tension is given by the weight of liquid displaced in the meniscus (light gray with dotted outline). (b) Two methods for estimating the drag-speed relationship for the triangular body and floater assembly. The dashed curve uses previous measurements of the drag coefficient for thin plates in parallel motion through a fluid. Solid curves represent inferences of speed V and drag D from position measurements of the assembly coasting to a stop after a "kick" impulse. The gray shaded region represents an error of 30%, and data from five trials are shown.

The bearing system allows for low-friction lateral motion over a range of several centimeters. This range is set by the length (8 cm) of the glass panes as compared to the length (2 cm) of the floaters, which permits displacements of no more than about $y = \pm 3$ cm lateral to the direction of wave propagation. In the wave propagation direction (x), the floaters are quite free within the panes, as the spacing between the panes is 7.5 cm and the associated width of the floaters is 3 cm. When waves are generated, the assembly tends to sit off-center in the x direction and somewhat closer to the downstream panes, i.e., toward the right in Figs. 2(b) and 2(d). This suggests that a force in the propagation direction is exerted on the body by the waves. No visible differences can be detected in the x positions of the two floaters, as would suggest a torque on the body.

The glass panes must be precisely aligned, as any angle between the panes in a pair would induce lateral forces due to the surface tension repulsion. We ensure alignment with a rigid bridge of glass that runs laterally over the tank and above the water and against which all four glass panes are placed. Control experiments show that the body-floater assembly may sit within the glass panes for tens of minutes, indicating negligibly weak lateral forces in the absence of waves.

F. Body geometrical parameters

The geometries are fabricated by laser cutting the acrylic plastic components and carefully gluing them to carbon fiber rods. Determining the small depth h at which the upper surfaces of the body-floater assembly sit below the free surface is more challenging. We use previous results regarding the vertical force balance of partially submerged bodies to infer the value of h from known and measurable quantities. Namely, Keller has given a generalization of Archimedes' principle for such bodies [52] whose depth is related to the balance of weight against both buoyancy and surface tension, as illustrated in Fig. 5(a). The buoyant force is shown to be given by the weight of the liquid displaced by the object, which is associated both with the volume of the solid itself (dark gray) and the volume directly above the solid up to the waterline (light-gray region outlined by dashed lines). The vertical component of surface tension is also given by such a weight of displaced liquid, namely, that displaced by the meniscus (gray region with dotted outline). Thus the body sits

at a depth such that its total weight (as measured "dry" in air) equals the sum of the weights of liquid associated with the volume displaced by the body itself, by the region directly above the body, and by the meniscus.

Further, the volume associated with the meniscus simplifies under the assumption of small deflections h, in which case the surface shape rises as $z \approx -h\exp(-r/\lambda_c)$ at a distance r from the contact point [53]. Here $\lambda_c = \sqrt{\gamma/\rho g}$ is the capillary length, which takes on a value of 2.7 mm for the air-water interface. The cross-sectional area of the meniscus on one side of the body is then $h\lambda_c$, and the total volume in the meniscus is $h\lambda_c L_f$, where L_f is the total circumferential or perimeter length of the body.

Under these assumptions, a solid object that has some portions fully submerged and others floating (i.e., breaching the surface) yields a vertical force balance whose expression can be rearranged to solve for the desired depth h:

$$Mg = \rho(V_s + hA_f + h\lambda_c L_f)g \;\rightarrow\; h = \frac{M/\rho - V_s}{A_f + \lambda_c L_f}. \tag{2}$$

Here the solid has total mass M and volume V_s, and the floating portions have total planform area A_f (when viewed from above) and circumferential length L_f. In the first expression, the three terms on the right-hand side represent displaced liquid weights for the solid itself, the region directly above the solid, and the meniscus. The geometry of the assemblies studied here are targeted toward a depth on the order of 1 mm, which has been shown to lead to significant refraction. The above formula and direct measurements of the mass and geometrical parameters yield values of $h = 0.96$ mm for the triangle, $h = 1.02$ mm for the disk used as a control in planar waves, $h = 1.07$ mm for the disk used in the trapping experiments, and $h = 1.16$ mm for the rectangle used in the plane wave visualizations. An estimate of the bending of the carbon fiber rods under the buoyancy-reduced weight of the body results in a maximal deflection of approximately 0.01–0.05 mm or 1%–5% of the depth h, which can be considered negligibly small.

G. Hydrodynamic drag estimates

Connecting the body speed measured in the transport experiments to theoretical estimates of the refractive force requires a drag-speed relationship for the triangular body assembly. We consider two ways to estimate this drag. First, we use previous measurements [54] of the drag on flat plates in parallel motion through a fluid at comparable Reynolds numbers, defined as $\mathrm{Re} = VL/\nu$ for a plate of length L moving at speed V through a fluid of kinematic viscosity ν. For the transport experiments to be reported here, the typical values of $L = 10$ cm, $V = 0.05$ cm/s, and $\nu = 10^{-2}$ cm^2/s (water) yield $\mathrm{Re} = 50$. At these intermediate Reynolds numbers, previous measurements indicate that the drag dependence on speed is well approximated by the law

$$D = \tfrac{1}{2}\rho V^2 C_D(\mathrm{Re})A_w \text{ with } C_D(\mathrm{Re}) = 4\mathrm{Re}^{-2/3} \text{ and } \mathrm{Re} = VL/\nu, \tag{3}$$

where ρ is the density of the fluid, A_w is the wetted area of the plate, and $C_D(\mathrm{Re})$ is the drag coefficient [54]. Inputting the parameter values $\rho = 1$ g/cm^3 and $A_w = 72$ cm^2 as relevant to our experiments, we arrive at the drag-speed relationship $D(V)$ plotted as the dashed curve in Fig. 5(b). Here we have taken the wetted area to be a two-sided area of the triangular body summed with the one-sided areas of the two floaters. The plotted region focuses on speeds of order 1 mm/s, as will prove relevant to our experimental observations.

These calculations should be viewed as order-of-magnitude estimates of the force given the differences between the conditions studied in the literature and those of our experiments. First, the literature pertains to rectangular plates, whereas our body is triangular. Further, the reported conditions are deep within a fluid, whereas our bodies are near the free surface. Third, the literature considers motion within an otherwise quiescent fluid, whereas body motions in our experiments are accompanied by waves and the associated fluid motions.

For these reasons, we also seek a more direct experimental measurement of the drag on the triangular assembly. Conventional techniques would involve connecting the object to a force gauge and holding within the flow of a water tunnel or pulling through a tow tank, for which care must be taken to capture the unique hydrodynamical environment and especially the proximity to the free surface [55]. As an alternative, we consider experiments in which the triangle-floater assembly is suspended as in the transport experiments and then given an impulsive "kick" that launches it forward (apex leading). The assembly then coasts to a stop under the action of fluid drag. Capturing the motion on video and tracking position over time then allows access to the instantaneous velocity (derivative of position) and force, which Newton's second law indicates is the assembly mass times the acceleration (second derivative of position). This force may be assumed to be the fluid drag. In Fig. 5(b) we plot as solid curves the drag-speed relationship obtained for five such perturbation experiments. While such experiments are contaminated by flow entrainment and a true steady speed is never reached, the resulting data show fair repeatability at the level of 30%, as indicated by the shaded band. The data also show acceptable agreement with the drag model (dashed curve) derived from the literature.

III. DEMONSTRATIONS AND RESULTS ON TRANSVERSE TRANSPORT AND TRAPPING

Employing the experimental techniques described in the previous section, we demonstrate here that refractive effects from surface waves can lead to forces and motions observable at the laboratory scale. We also show how these forces can be used to transversely move and position submerged bodies. In this section we first document observations of the transport of asymmetric bodies under the action of plane waves. We then present simplified force estimates based on the theory of radiation stresses, and we show that the observed motions are consistent in magnitude with such forces. Finally, the ideas are extended to achieve the trapping of symmetric bodies due to interactions with focused wave fields.

A. Transverse transport of asymmetric bodies in plane waves

The systems described in the previous section permit the direct experimental testing of a scheme for harnessing planar waves to drive motions transverse to the direction of wave propagation. Firstly, we use the outlined methods to confirm that significant refraction is achieved for the typical conditions achievable in experiments. A body of rectangular planform shape serves well for this purpose, and shadowgraph imaging produces the photo of Fig. 6(a). The sketch of panel (a′) highlights the key features of interest. Refraction over the body is seen as kinks in the wave fronts that relate to the shorter wavelength and slower wave speed over the body as compared to the bulk. The observed length or speed ratio of about 2 is substantial. Further, careful inspection of the shadowgraph just downstream of the rectangle shows that the image gradients are lessened relative to the upstream field, which is consistent with the measured decrease in wave amplitude. Coming to the asymmetric geometry of interest, Figs. 6(b) and 6(b′) show analogous results for the case of a triangular body. Kinked wave fronts, shorter wavelengths, and relatively muted gradients in the downstream are similarly seen. The most prominent signature of refraction of interest here is the deflection of the downstream wave fronts, indicating that the emitted wave is redirected with a transverse velocity away from the apex of the triangle. This is expected based on the optical analog of a prism.

When released within the capillary bearing system, triangular objects are observed to undergo motion toward their apex and oppositely from the direction of propagation of the refracted waves. The data of Fig. 6(c) show three trials each of the transverse position coordinate y versus time for bodies aimed rightward (red) as viewed from the wavemaker, which are observed to move rightward (decreasing y), and those aimed leftward (blue), which move leftward (increasing y). Since the zero point of time is arbitrary, here we choose values that ease the comparison across trials. Errors in the position data are comparable to or smaller than the line thicknesses. Videos of these motions are

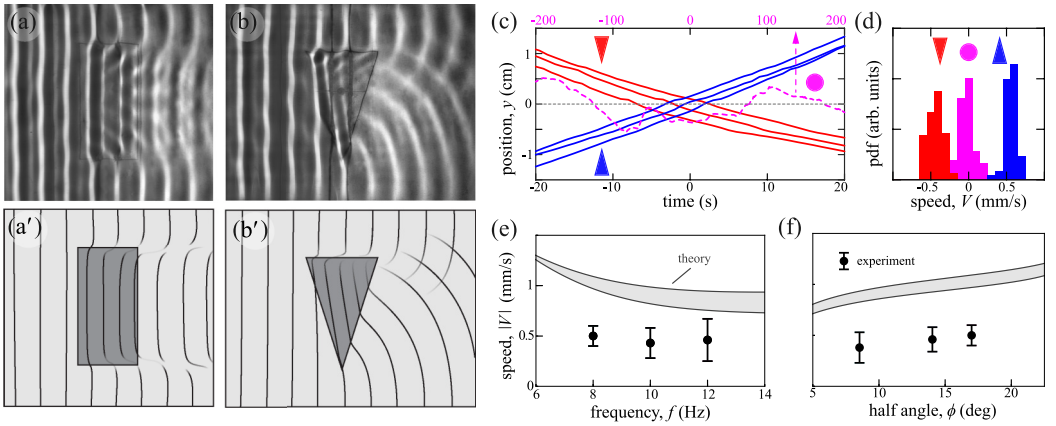


FIG. 6. Transverse transport of an asymmetric body in plane waves. (a) and (a′) Shadowgraph and schematic for a submerged rectangular body (length 10 cm) with waves traveling left to right. In the shadowgraph image, wave crests appear light and troughs dark. Refraction induces shorter wavelength over the body, and the slower wave speed leads to kinked wave fronts in the wake. (b) and (b′) A triangular body (length 10 cm) acts as a prism that redirects oncoming waves. (c) Experimental measurements of transverse position over time show apex leading motions. Shown are three trials each of bodies aimed rightward (red) as viewed from the wavemaker and leftward (blue). A control study of a symmetric, circular body (magenta) reveals slow and undirected motion (note time on upper axis). (d) Speed statistics for the data of (c). Parameters: triangle length $L = 10$ cm at depth $h = 1$ mm, disk diameter of 6 cm at depth 1 mm, wave frequency $f = 8$ Hz and incoming amplitude $A = 0.5$ mm. (e) and (f) Average speeds for experiments with varying wave frequency f and body half angle ϕ. Error bars represent standard deviations over three independent trials. The black curves represent theoretical estimates of the maximum speed from Eq. (4) translated through the expression $C_D(\mathrm{Re})$, and the shaded bands represent the variations due to surface tension attenuation.

available as Supplemental Material (SI) [56]. In all cases, it is seen that the body moves with its apex leading and with relatively uniform progression over time. We interpret the opposite signs for the speeds in the two sets of experiments as strong evidence that the direction of travel is dictated by the body geometry and not other aspects of the experimental system that may induce asymmetric forcing.

As a control, we conduct additional experiments involving plane waves sent over a thin disk, and the measured position dynamics for this symmetric body is shown as the magenta dashed curve of Fig. 6(c). Note that for this case we display time on a 10-times compressed scale labeled on the upper axis, allowing the much slower dynamics to be presented on the same plot. This symmetric body wanders about with low speed and without preferred direction. These findings provide further evidence that body asymmetries are essential to achieving persistent directed motion. The control experiments can also be interpreted as providing the scale of speed fluctuations that arise from unintended perturbations that may be present in the experimental system such as flows in the tank, surface tension forcing from the bearing, or wave and flow effects on the floaters.

The measurements are further analyzed to produce the speed histograms in Fig. 6(d), where data across trials are pooled together. When aimed rightward (red) as viewed from the wavemaker, triangular bodies display speeds of mean and standard deviation 0.44 ± 0.10 mm/s, and leftward (blue) yields 0.56 ± 0.08 mm/s. The disk displays speeds of 0.016 ± 0.11 mm/s, and the significantly lower mean is consistent with the expected lack of directed forcing. The difference in means between the right- and left-going cases for triangles is at the 20% level and is therefore comparable to the variations in speed for any given condition, suggesting that the left- and right-going cases are not significantly different. Given the uniformly small measurement errors, the spreads seen in the

data of Fig. 6(d) should be understood as actual speed variations. Taken together, these encouraging observations motivate a quantitative comparison to theory, which is pursued in what follows.

B. Theoretical estimate of refractive force on a submerged triangular body

Here we present a simplified theory based on the geometric optics of a prism and which yields quantitative estimates for the refractive force on a triangular object interacting with water waves that pass over it. The incoming waves represent a flux of momentum that is intercepted by the body and redirected by refraction in the shallow layer, and the resulting change in lateral momentum is associated with a transverse force on the object. We first consider the radiation stress S in a wave field as computed in the classical work of Longuet-Higgins and Stewart for the case of plane gravocapillary traveling waves on the surface of a fluid of uniform depth [21]. We then suppose that a submerged object intercepts the waves impinging across its lateral extent L and that the waves are refractively redirected so as to be emitted at an angle θ and with diminished amplitude due to damping during transit over the body. Then the principle of action and reaction implies a corresponding force imparted to the object:

$$F = \left(\frac{a}{A}\right)^2 SL\sin\theta, \text{ where } S = \frac{1}{2}\rho g A^2(1+\epsilon)\left[\frac{2kH}{\sinh 2kH} + \frac{1+3\epsilon}{2(1+\epsilon)}\right]. \tag{4}$$

Here S is the time-averaged and depth-integrated momentum flux associated with the incoming plane waves and whose expression includes the gravitational acceleration constant g, the fluid density ρ, and the capillarity factor $\epsilon = \gamma k^2/\rho g$ that involves the surface tension coefficient γ. The other parameters are the wave number k, the incoming amplitude A, the outgoing amplitude $a < A$, and the depth H of the bulk.

To specify the refraction angle θ, we invoke Snell's law and the relative phase speeds of waves in the shallow layer over the body as compared to the bulk. The phase speed for gravocapillary waves obeys the relation $c_p = \omega/k = \sqrt{(g/k)(1+\epsilon)\tanh kd}$, where the dependency on the depth d is explicit and also implicit through the wave number k via the dispersion relation Eq. (1). We define the higher speed of waves over the bulk as $U = c_p(d = H)$ and the lower speed in the shallow layer as $u = c_p(d = h)$. Snell's law can then be applied successively at the two sloped sides of a triangular body of half-angle ϕ to arrive at the total deflection angle

$$\theta = \arcsin\left[n\sin\left(2\phi - \arcsin\left(\frac{1}{n}\sin\phi\right)\right)\right] - \phi \text{ where } n = \frac{U}{u} \tag{5}$$

of the outgoing waves relative to the incoming. Here $n > 1$ is the relative index of refraction.

The formulas of Eqs. (4) and (5) constitute force predictions as related to known and measurable quantities about the wave field and body geometry. The assumptions of the theory suggest that the results ought to be viewed as indicative of the maximum available force associated with wave refraction and an overestimate of what might be attained in experiments. Firstly, refraction alone is considered and not other aspects such as reflection and diffraction that are present in the more general scattering problem. Moreover, the refraction analysis is idealized by ignoring the fact that some portion of the waves refracted at the first sloped edge reach the base of the triangle rather than the second sloped edge. Further, we have assumed that the momentum flux integrated over the entire bulk depth to be redirected, whereas only the portion associated with the shallow layer seems relevant. Nonetheless, the theory can serve as a useful point of comparison to experiments, which necessitates converting the predicted refractive force F to an expected speed V. To do so, we assume the force balance $F = D(V)$ and employ the drag-speed relation provided in Sec. II G.

C. Comparison of theory and experiment for transverse transport

We first assess the case relevant to the images and data of Figs. 6(b)–6(d), i.e., a triangle of base 6 cm and height 10 cm submerged a depth $h = 0.96$ mm below the surface and subject to

plane waves of frequency $f = 8$ Hz. The predicted phase speeds over the bulk and the body are $U = 23$ cm/s and $u = 12$ cm/s respectively, and the predicted index of refraction of $n = U/u \approx 2$ is consistent with the degree of shortening of the wavelength seen in the shadowgraph images of (b) and (b′). Moreover, the theory predicts a deflection angle of $\theta = 0.5 \approx 30^\circ$, which is qualitatively corroborated by the images.

To assess the speed, we pool the data for the right- and left-moving cases by evaluating the absolute value $|V|$, and the resulting mean and standard deviation are shown as the leftmost marker and vertical bars in panel (e). This is to be compared to the theory curve above, which is computed from the formulas of Sec. III B for the relevant parameter values. It is seen that the measured speed is about 40% that predicted by theory, and the deviation is significantly larger than what could be explained by fluctuations in the measurements. These findings are consistent with the view given above that the theory is an overestimate due to its idealized assumptions. Additionally, it should be noted again that impurities tend to lower the surface tension coefficient γ by as much as 50% [50]. A sensitivity analysis of the model equations indicates consequent decreases in the predicted force that are displayed as the shaded band in Fig. 6(e). The effect is of order 10%–20% and partially but not dominantly explains the observed discrepancy between theory and experiment.

To provide further comparisons, we conduct additional experiments for varying frequency f and apex half angle ϕ of the triangle. The plot of Fig. 6(e) represents the body geometry considered previously with $\phi = 17^\circ$ subject to waves of different f. The plot of panel (f) represents fixed $f = 8$ Hz for different apex angles, as achieved by keeping the triangle height fixed at 10 cm and varying the base length through the values 3, 5, and 6 cm. Again it is seen that the theory (gray band) predicts forces of the right order but overestimated in all cases, and fairly consistently by about a factor of 2. Moreover, the theory and experiments display similar trends with the quantities, which are in both cases weak over the ranges that can be conveniently accessed in experiments. Namely, the force and speed seem to slightly decrease with f and increase with ϕ, though any such trend is comparable to the observed variations. Despite this, these findings suggest that some degree of control over the motion may be achieved by varying the wave parameters and body geometry.

D. Trapping of a symmetric body in focused waves

We next aim to use wave-induced forces for trapping or holding a submerged body in place at a desired location. The scheme proposed here involves focused water waves passing over a symmetric body. We consider a disk as a simple shape that is suspended within the wave field emitted by a circular-arc wavemaker vibrating at the free surface. The waves ought to increase in amplitude as they converge on the focal point, i.e., the geometric center of the arc, and therefore transverse gradients in wave amplitude are expected. The shadowgraph image of Fig. 7(a) confirms as much, where the submerged disk can be seen near the focus. The panel (a′) shows an interpretative schematic. As expected, the downstream wave field is laterally symmetric when the disk is centered in the trap. The case of Figs. 7(b) and 7(b′) shows the response when the body is transversely displaced (here, leftward as viewed from the wave source), from which it is seen that the downfield waves are deflected in the same direction. These findings are suggestive of a force of the same nature as that in the transport experiments and which is so directed as to return the body to the focal point.

We therefore carry out experiments in which the body is free to translate transversely due to the linear bearing system described previously. When vibrations of the wavemaker are initiated, it is observed that the body drifts toward the focal point, where it thereafter remains remarkably fixed, typically to within a few millimeters. To demonstrate this trapping effect, we perturb the body with a brief lateral impulse and measure the resulting transverse position y over time. The corresponding measurements are displayed in Fig. 7(c), from which it is seen that the disk robustly returns to the focal point $y = 0$ when perturbed in either direction and for a range of strengths. Associated videos are available as Supplemental Material [56]. The typical speeds during the recovery phase are of the same order as those observed in the transport experiments, suggesting a similar origin for the forces.

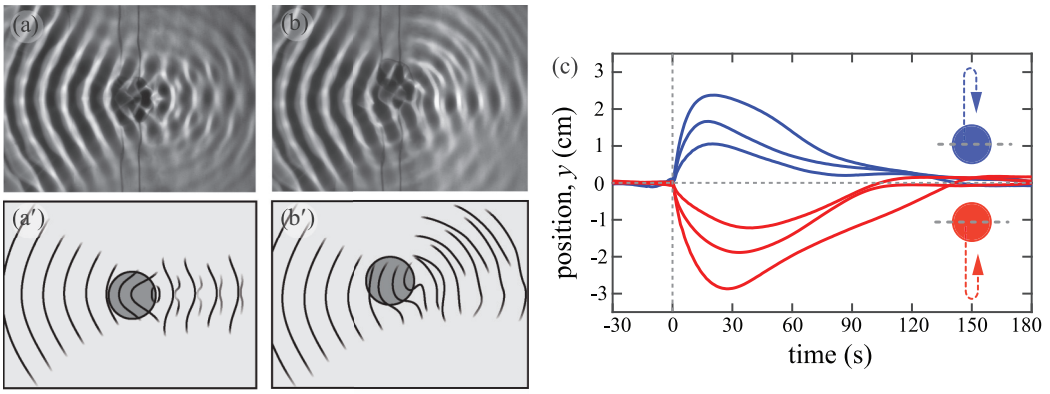


FIG. 7. Trapping of a symmetric body in focused waves. (a) and (a′) A circular body (diameter 4 cm) located near the focal point of converging waves. (b) and (b′) If perturbed to one side of the focal point, the waves are redirected to the same side. (c) Measurements of transverse position show robust trapping. Shown are three trials each of bodies perturbed rightward or in the $-y$ direction (red) and leftward ($+y$, blue) by an impulse applied at time zero.

These experiments show that focused waves may be used as "hydrodynamic tweezers" for laterally confining objects at specified locations. The robust return of the body to the focal center can be explained by again associating wave deflection in one direction with a reaction force on the body in the opposite direction. Thus this tweezing action exploits gradients in wave amplitude along the lateral direction, and the body seeks locations of high amplitude. The shadowgraph images of Figs. 7(a) and 7(b) are consistent with this picture while also making clear that the wave field is complex. A quantitative explanation for the observed dynamics may therefore require consideration of the full scattering problem and perhaps direct numerical simulations of the flow coupled to the free surface motion.

IV. DISCUSSION AND CONCLUSIONS

The findings reported here advance the study of wave-structure interactions by experimentally showing that the forces associated with refracting water waves lead to demonstrable effects at the laboratory scale. Namely, we show that submerged objects of asymmetric planform shape can be induced to move laterally when subjected to plane gravocapillary waves, and symmetric objects can be trapped within focused waves. These phenomena are made observable for the first time via a suspension and bearing system that uses surface tension to both hold an object just beneath the air-water interface as well as allow for its low-friction linear motion, isolated along a chosen axis. These specialized techniques prove to be adequately sensitive for detecting responses to the submilligram-scale forces generated by low-amplitude waves typical of ripple tank laboratory setups. Further, we provide a simplified mechanistic picture based on the geometric optics of refraction together with estimates of the radiation stresses that furnish theoretical predictions of the forces and motions. Comparisons to experimental data show fair agreement at the level of a factor of 2, which might be expected given the idealizations underlying the theoretical estimates. Encouragingly, theory and experiments display similar trends with regard to responses for varying body geometries and wave parameters in the case of the transverse transport of triangular bodies in plane waves. Further, experiments demonstrate a robust tweezing or trapping effect in which objects are attracted to points of high wave amplitude, and the observed motions and wave fields are consistent with refraction-based forces. The transport and trapping behaviors documented here have parallels to optical and acoustic effects, and hence this work adds to a growing list of studies involving water waves that exploit analogies with other forms of waves [39–42].

As but the first report on the dynamical response of submerged and mobile structures due to wave refraction, there remains much to do in future work. Experimentally, there are parametric sweeps that ought to be conducted beyond the limited explorations of triangle apex angle and wave frequency provided here. The body depth below the interface seems particularly interesting in that it controls refraction but likely also the degree of damping of the transmitted waves, and the two effects would seem to impact refractive forces with oppositely trending dependencies on depth. It may also be worthwhile to conduct direct force measurements on fixed structures, and the estimates outlined here can guide the experimental design by providing estimates of the expected magnitudes. While our focus here is on the subtle effects of transverse forces and our experimental systems isolate such effects, future studies should look at the more direct and likely larger-scale effects in the direction of wave propagation. More generally, experimental techniques for extracting the global wave field could be useful for directly measuring momentum flux [57–59] but would also require careful validation for use in settings involving wave-structure interactions. Further afield, it may be interesting to introduce multiple free bodies that may interact through their scattering fields. On the theory side, we have given an idealized and approximate picture of the dominant physics, and a more complete theory would be challenged to account for the full scattering problem, including reflection as well as the viscous effects that cause wave damping over the body [26,27]. Another approach might be to solve the complete fluid dynamical problem via direct numerical simulations of the full Navier-Stokes equations [60–62]. Some major challenges include the wide range of scales, sharp-edged geometries, mobility of the solid boundary, and the free interface. More broadly and across experimental, computational, and modeling approaches, future studies may fruitfully look to other wave systems for inspiring problems and objectives, for example, achieving multibody control and positioning via holographic tweezers [63,64].

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DATA AVAILABILITY

The data that support the findings of this article are included in the article.

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